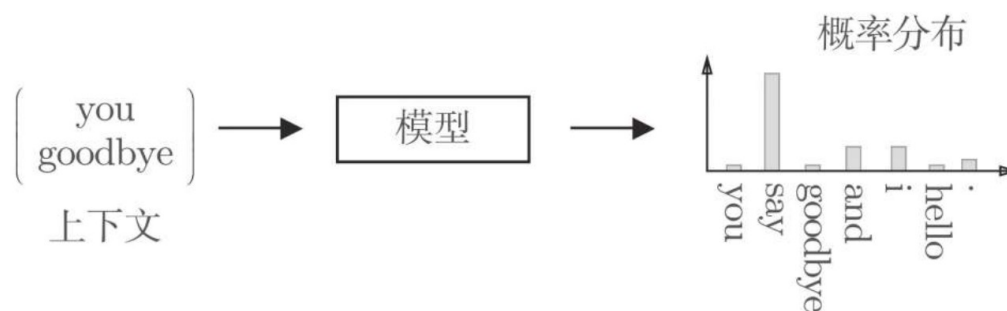


GloVe词向量

GloVe: Global Vectors for Word Representation

词向量方法分类-分布式假设

- 基于计数的方法：
 - 使用整个语料库的统计数据(共现矩阵)
 - SVD
- 基于推理的方法：
 - 输入某种模型，模型接收上下文信息作为输入，并输出（可能出现的）各个单词的出现概率
 - 使用语料库来学习模型，使之能做出正确的预测
 - 作为模型学习的产物，得到单词的分布式表示



词向量方法分类-分布式假设

- 基于计数的方法：
 - 例：奇异值分解SVD
 - 缺点：处理大规模语料库(稀疏性、计算量)
- 基于推理的方法：
 - 例：word2vector(CBOW、skip-gram)
 - 缺点：只能利用一定窗长的上下文环境，即利用局部信息，没法利用整个语料库的全局信息

两者是否可以结合？

GloVe

词的独热表示

我 喜欢 自然 语言 处理。
我 爱 深度 学习。
我 喜欢 机器 学习。



词表: “我” “喜欢” , “自然” “语言” , “处理” “爱” ,
“深度” “学习” , “机器” , “。”

词表大小 : 10



- 词语共现频次表
 - 每一项代表一个词 w_i 与另一个词 w_j (上下文)在同一个句子中的共现频次，每个词与自身的共现频次设置为0

	我	喜欢	自然	语言	处理	爱	深度	学习	机器	。
我	0	2	1	1	1	1	1	2	1	3
喜欢	2	0	1	1	1	0	0	1	1	2
自然	1	1	0	1	1	0	0	0	0	1
语言	1	1	1	0	1	1	0	0	0	1
处理	1	1	1	1	0	0	0	0	0	1
爱	1	0	0	0	0	0	1	1	0	1
深度	1	0	0	0	0	1	0	1	0	1
学习	2	1	0	0	0	1	1	0	1	1
机器	1	1	0	0	0	0	0	1	0	1
。	3	2	1	1	1	1	1	2	1	0

词的分布式表示

我 喜欢 自然 语言 处理。
我 爱 深度 学习。
我 喜欢 机器 学习。

	我	喜欢	自然	语言	处理	爱	深度	学习	机器	。
我	0	2	1	1	1	1	1	2	1	3
喜欢	2	0	1	1	1	0	0	1	1	2
自然	1	1	0	1	1	0	0	0	0	1
语言	1	1	1	0	1	1	0	0	0	1
处理	1	1	1	1	0	0	0	0	0	1
爱	1	0	0	0	0	0	1	1	0	1
深度	1	0	0	0	0	1	0	1	0	1
学习	2	1	0	0	0	1	1	0	1	1
机器	1	1	0	0	0	0	0	1	0	1
。	3	2	1	1	1	1	1	2	1	0

共现矩阵它有以下3个特点：

- 统计的是单词对在给定环境中的共现次数；在一定程度上能表达词间的关系
- 共现频次计数是针对整个语料库而不是一句或一段文档，具有全局统计特征
- 共现矩阵它是对称的

词的分布式表示——共现概率

共现矩阵

□ 元素 x_{ij} : 词 j 出现在词 i 的上下文中的次数

□ 共现矩阵行和 $x_i : x_i = \sum_k x_{ik}$

□ 出现在词 i 的上下文中的所有词出现总次数

□ 共现概率: 词 j 出现在词 i 的上下文中的概率 $P_{ij} = P(j|i) = \frac{x_{ij}}{x_i}$

	我	喜欢	自然	语言	处理	爱	深度	学习	机器	。
我	0	2	1	1	1	1	1	2	1	3
喜欢	2	0	1	1	1	0	0	1	1	2
自然	1	1	0	1	1	0	0	0	0	1
语言	1	1	1	0	1	1	0	0	0	1
处理	1	1	1	1	0	0	0	0	0	1
爱	1	0	0	0	0	0	1	1	0	1
深度	1	0	0	0	0	1	0	1	0	1
学习	2	1	0	0	0	1	1	0	1	1
机器	1	1	0	0	0	0	0	1	0	1
。	3	2	1	1	1	1	1	2	1	0

- how **certain aspects of meaning** can be extracted directly from co-occurrence probabilities
- Consider **two words i and j** that exhibit **a particular aspect of interest**
- suppose we are interested in the concept of thermodynamic phase for which we might take **$i = \text{ice}$ and $j = \text{steam}$**
- The **relationship** of these words can be examined by **studying the ratio of** their co-occurrence probabilities with various probe words, k .

$i = \text{ice}$ and $j = \text{steam}$

- For words k related to ice but not steam, say $k = \text{solid}$, we expect the ratio P_{ik}/P_{jk} will be large.
- for words k related to steam but not ice, say $k = \text{gas}$, the ratio should be small.
- For words k like water or fashion, that are either related to both ice and steam, or to neither, the ratio should be close to one.

i= ice j= steam	K的取值	与i、j的相关性	x_{ik}/x_{jk}
	K=solid	与i相关, j不相关	大
	K=gas	与j相关, i不相关	小
	K=water	与j、i均相关	≈ 1
	K=fashion	与j、i均不相关	≈ 1

□ Co-occurrence probabilities for target words ice and steam with selected context words from a 6 billion token corpus.

Probability and Ratio	$k = solid$	$k = gas$	$k = water$	$k = fashion$
$P(k ice)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k steam)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k ice)/P(k steam)$	8.9	8.5×10^{-2}	1.36	0.96

□ Compared to the raw probabilities, the ratio is better able to distinguish relevant words (solid and gas) from irrelevant words (water and fashion) and it is also better able to discriminate between the two relevant words

GloVe模型

- The above argument suggests that the appropriate starting point for word vector learning should be with **ratios of co-occurrence probabilities** rather than **the probabilities themselves**.
- Noting that the ratio P_{ik}/P_{jk} depends on three words i , j , and k , the most general model takes the form

$$F(w_i, w_j, \tilde{w}_k) = \frac{P_{ik}}{P_{jk}}$$

- where $w \in \mathbb{R}^d$ are word vectors and $\tilde{w} \in \mathbb{R}^d$ are separate context word vectors
 - The right-hand side : is extracted from the corpus
 - F may depend on some unspecified parameters
- ? 问题 回归

函数F的选择

① F encode the information present the ratio P_{ik}/P_{jk} in the word vector space

- vector spaces are inherently linear structures the most natural way to do this is with vector differences

$$F(w_i, w_j, \tilde{w}_k) = \frac{P_{ik}}{P_{jk}} \longrightarrow F(w_i - w_j, \tilde{w}_k) = \frac{P_{ik}}{P_{jk}}$$

② 右侧结果是个标量，左边是个关于向量的函数

- F could be a complicated function parameterized by, e.g., a neural network, doing so will obfuscate the linear structure trying to capture
- To avoid this issue, take the dot product of the arguments

$$F((w_i - w_j)^T \tilde{w}_k) = \frac{P_{ik}}{P_{jk}}$$

- preventing F from mixing the vector dimensions in undesirable ways

- the distinction between a word and a context word is arbitrary and that we are free to exchange the two roles
- we must not only exchange $w \leftrightarrow \tilde{w}$, but also $X \leftrightarrow X^T$
- ① we require that F be a homomorphism between the groups $(\mathbb{R}, +)$ and $(\mathbb{R}_{>0}, \times)$, i.e

$$F((w_i - w_j)^T \tilde{w}_k) = \frac{P_{ik}}{P_{jk}} \quad F((w_i - w_j)^T \tilde{w}_k) = \frac{F(w_i^T \tilde{w}_k)}{F(w_j^T \tilde{w}_k)}$$

$$F(w_i^T \tilde{w}_k) = P_{ik} = \frac{X_{ik}}{X_i}$$

$F = \exp$

$$w_i^T \tilde{w}_k = \log(P_{ik}) = \log(X_{ik}) - \log(X_i) .$$

GloVe模型

- ② $\log(X_i)$ is independent of k and can be absorbed into a bias b_i for w_i

■ an additional bias $\tilde{b}_k$ for $\tilde{w}$ restores the symmetry

$$w_i^T \tilde{w}_k = \log(P_{ik}) = \log(X_{ik}) - \log(X_i) . \longrightarrow w_i^T \tilde{w}_k + b_i + \tilde{b}_k = \log(X_{ik}) .$$

- ③ **Diverges** whenever its argument is zero

$$\log(X_{ik}) \rightarrow \log(1 + X_{ik}) ,$$

- ④ **drawback**: weight all co-occurrences equally

- rare cooccurrences are noisy and carry less information than the more frequent ones
- yet even just the zero entries account for 75–95% of the data in X depending on the vocabulary size and corpus

- propose a new weighted least squares regression model

$$J = \sum_{i,j=1}^V f(X_{ij}) \left(w_i^T \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

- The weighting function should obey the following properties
 1. $f(0) = 0$. If f is viewed as a continuous function, it should vanish as $x \rightarrow 0$ fast enough that the $\lim_{x \rightarrow 0} f(x) \log^2 x$ is finite.
 2. $f(x)$ should be non-decreasing so that rare co-occurrences are not overweighted.
 3. $f(x)$ should be relatively small for large values of x , so that frequent co-occurrences are not overweighted.

$$f(x) = \begin{cases} (x/x_{\max})^\alpha & \text{if } x < x_{\max} \\ 1 & \text{otherwise} \end{cases}.$$

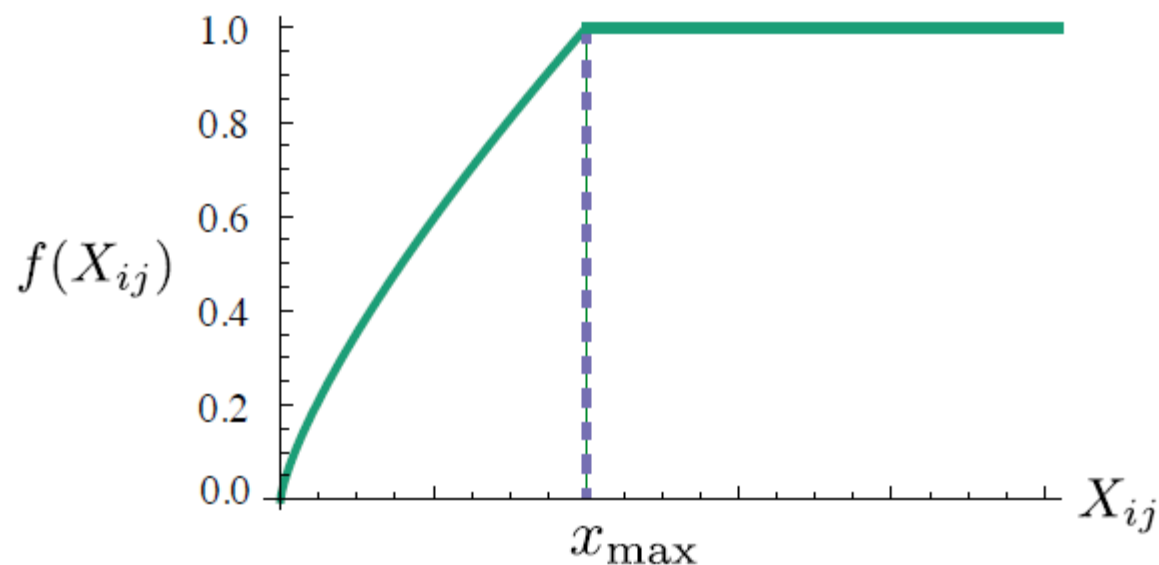


Figure 1: Weighting function f with $\alpha = 3/4$.

Glove模型训练过程：

- ① 从共现矩阵中随机采集一批非零词对作为一个mini-batch的训练数据；
- ② 随机初始化这些训练数据的词向量以及随机初始化两个偏置；
- ③ 然后进行内积和平移操作并与 $\log(x_{ij})$ 计算损失值，计算梯度；
- ④ 然后反向传播更新词向量和两个偏置；
- ⑤ 循环以上过程直到结束条件。

GloVe模型-实验结果

- 词的类比性
- 数据集大小：19,544
- 语义子集
- 语法子集

Table 2: Results on the word analogy task, given as percent accuracy. Underlined scores are best within groups of similarly-sized models; bold scores are best overall. HPCA vectors are publicly available²; (i)vLBL results are from (Mnih et al., 2013); skip-gram (SG) and CBOW results are from (Mikolov et al., 2013a,b); we trained SG[†] and CBOW[†] using the word2vec tool³. See text for details and a description of the SVD models.

Model	Dim.	Size	Sem.	Syn.	Tot.
ivLBL	100	1.5B	55.9	50.1	53.2
HPCA	100	1.6B	4.2	16.4	10.8
GloVe	100	1.6B	<u>67.5</u>	<u>54.3</u>	<u>60.3</u>
SG	300	1B	61	61	61
CBOW	300	1.6B	16.1	52.6	36.1
vLBL	300	1.5B	54.2	<u>64.8</u>	60.0
ivLBL	300	1.5B	65.2	63.0	64.0
GloVe	300	1.6B	<u>80.8</u>	61.5	<u>70.3</u>
SVD	300	6B	6.3	8.1	7.3
SVD-S	300	6B	36.7	46.6	42.1
SVD-L	300	6B	56.6	63.0	60.1
CBOW [†]	300	6B	63.6	<u>67.4</u>	65.7
SG [†]	300	6B	73.0	66.0	69.1
GloVe	300	6B	<u>77.4</u>	67.0	<u>71.7</u>
CBOW	1000	6B	57.3	68.9	63.7
SG	1000	6B	66.1	65.1	65.6
SVD-L	300	42B	38.4	58.2	49.2
GloVe	300	42B	81.9	69.3	75.0

GloVe测试结果

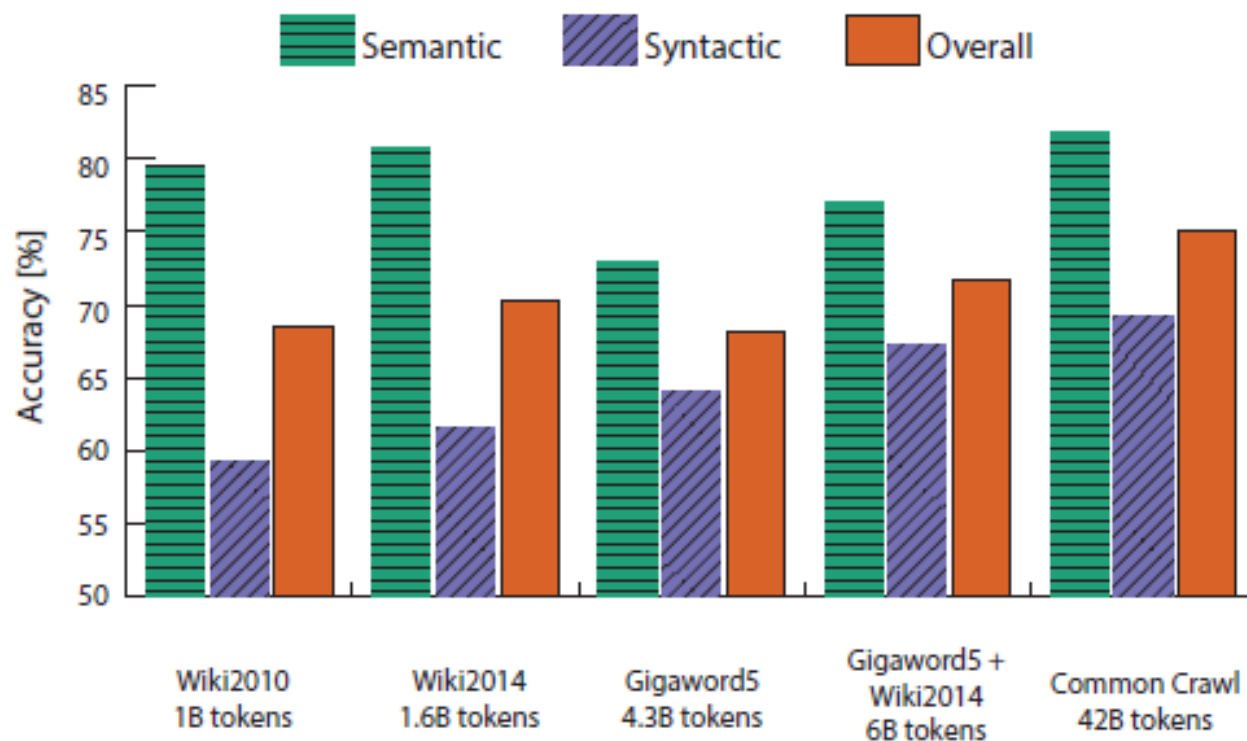
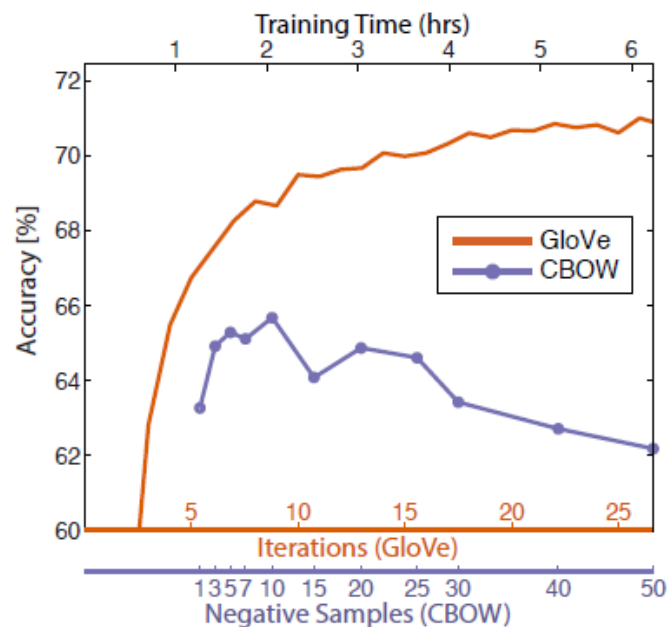
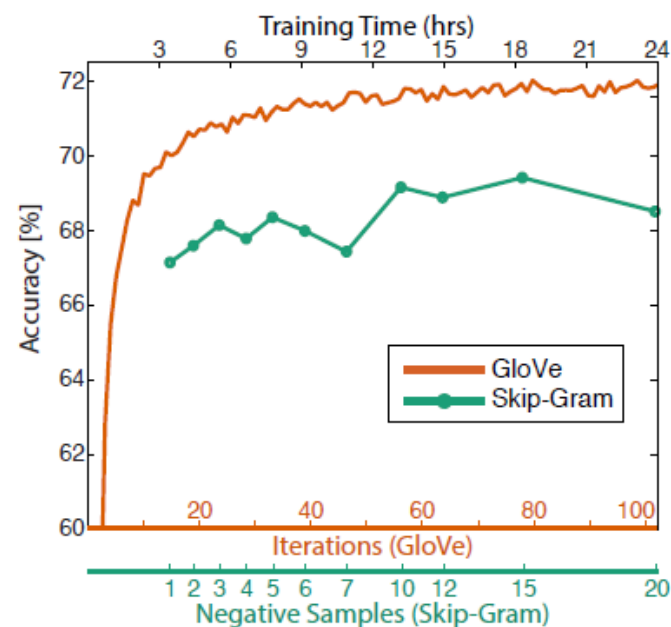


Figure 3: Accuracy on the analogy task for 300-dimensional vectors trained on different corpora.

GloVe测试结果



(a) GloVe vs CBOW



(b) GloVe vs Skip-Gram

Figure 4: Overall accuracy on the word analogy task as a function of training time, which is governed by the number of iterations for GloVe and by the number of negative samples for CBOW (a) and skip-gram (b). In all cases, we train 300-dimensional vectors on the same 6B token corpus (Wikipedia 2014 + Gigaword 5) with the same 400,000 word vocabulary, and use a symmetric context window of size 10.

GloVe模型-实验结果

- 词的相似性
- 数据集：见论文

Table 3: Spearman rank correlation on word similarity tasks. All vectors are 300-dimensional. The CBOW* vectors are from the `word2vec` website and differ in that they contain phrase vectors.

Model	Size	WS353	MC	RG	SCWS	RW
SVD	6B	35.3	35.1	42.5	38.3	25.6
SVD-S	6B	56.5	71.5	71.0	53.6	34.7
SVD-L	6B	65.7	<u>72.7</u>	75.1	56.5	37.0
CBOW [†]	6B	57.2	65.6	68.2	57.0	32.5
SG [†]	6B	62.8	65.2	69.7	<u>58.1</u>	37.2
GloVe	6B	<u>65.8</u>	<u>72.7</u>	<u>77.8</u>	53.9	<u>38.1</u>
SVD-L	42B	74.0	76.4	74.1	58.3	39.9
GloVe	42B	<u>75.9</u>	<u>83.6</u>	<u>82.9</u>	<u>59.6</u>	<u>47.8</u>
CBOW*	100B	68.4	79.6	75.4	59.4	45.5

Thanks!
Q&A ?

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